Name: _____

YEAR 11 SPECIALIST MATHS INVESTIGATIVE TASK 3

Proofs

Prerequisites

- Terminology such as converse, contrapositive and counter examples
- Proof by contradiction
- Proof by counter example
- Proof by mathematical reasoning (mainly numeric/algebraic)
- Proof using mathematical induction

[Total: 80 marks]

SECTION ONE

1 **a** If a statement of theorem says '*If P then Q*' then what do the converse and the contrapositive say?

b A famous French philosopher Descartes made the famous statement:

"*Je pense donc je suis*" i.e. "*I think, therefore I am.*"

i Write down the converse of Descartes' statement. Do you think it is valid?

ii Write down the contrapositive of Descartes' statement. If Descartes' statement is true, is the contrapositive valid?

c Consider the conjecture

"*If n^2 is an even integer then n is also an even integer.*"

i Write down the contrapositive.

ii By letting an odd integer be $2k + 1$, algebraically prove the contrapositive.

[2 + 3 + 5 = 10 marks]

2 Find a counter example to prove that the following statements are invalid.

i "If x and y are real numbers where $x > y$ then $x^2 > y^2$."

ii "All prime numbers are odd numbers."

iii "If p is a prime number, then $p + 2$ is also a prime number."

[3 marks]

3 It has been conjectured that:

The sum of four consecutive odd integers is divisible by 8.

Prove algebraically that the conjecture is valid. **[5 marks]**

4 A syllogism is a logical argument in which given two (or more) statements then a logical conclusion can be formed.

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For example:

- All Australians are human.
- Humans are mortal.
- Therefore Australians are mortal.

Consider the following statements:

- All squares are rectangles.
- All rectangles are quadrilaterals.
- All rectangles are parallelograms.
- All rhombuses are parallelograms
- All squares are rhombuses.
- All parallelograms are quadrilaterals.
- All squares are parallelograms.

a Find three statements from the list that form a syllogistic argument.

Write down the argument.

b Use this conclusion and another statement to make a further deduction.

c Use a Venn diagram to illustrate the relationship between quadrilaterals, parallelograms, squares, rectangles and rhombi.

Let Q represent quadrilaterals; P parallelograms; R rectangles; H rhombuses and S squares.

[2 + 2 + 4 = 8 marks]

5 **a** What can you deduce about the answer when

i an odd number is multiplied by an odd number?

ii an even number is multiplied by an even number?

b Prove your deductions for part **a**.

c Hence use a proof by contradiction to show that $\log_3^4$ is irrational.

[2 + 6 + 8 = 16 marks]

6 Prove using mathematical induction that $1 + 3 + 5 + 7 + \ldots + (2n - 1) = n^2$ for all natural numbers n.

[8 marks]

7 **a** Prove using mathematical induction that $(a - 1)$ will always be a factor of $a^n - 1$.

HINT: At one stage you will need to subtract a and add a to an expression.

You should also note that if a^n number is a factor of another number, then that number can be expressed as an integer multiplied by the factor.

(If p is a factor of q, then $q = mp$ where m is an integer.)

b A mathematician proposes a conjecture that $(a + 1)$ will always be a factor of $a^{2n+1} - 1$ if n is any non-negative integer and $a > 1$. As a first step he shows that if his conjecture works for $n = k$, it will work for $n = k + 1$.

i Show that if his conjecture works for $n = k$, it will work for $n = k + 1$.

ii Is this sufficient to prove the conjecture? Why, or why not?

iii What is the easiest way to prove his conjecture false? Prove the conjecture is, in fact, false.

iv Another mathematician proposes that $(a + 1)$ will NEVER be a factor $a^{2n+1} - 1$. Either prove this conjecture true, or find an example to show it is false.

HINT: If you are using proof by induction, start your proof with $n = 0$.

Note that if a number is not a factor of another number, then it will leave a remainder.
(If p is not a factor of q, then $q = mp + r$ where m is an integer and the remainder r must be at least 1 and less than p.)

Also, remember the trick of subtracting and adding the same variable to an expression!

v A corollary to a proof is something that logically follows from the proof. Can you see a corollary to your proof in part **iv**?

[10 + 18 = 28 marks]

8 A fallacious proof is one that includes an invalid line that is the cause of an invalid conclusion.

Find the invalid step in the following fallacious proof and explain carefully why the step is invalid.

Proof that 2 = 1

Let	$x = y$
Multiply both sides by	$x.\ x^2 = xy$
Add x^2 to both sides.	$x^2 + x^2 = x^2 + xy$
	$2x^2 = x^2 + xy$
Subtract $2xy$ from both sides.	$2x^2 - 2xy = x^2 + xy - 2xy$
	$2x^2 - 2xy = x^2 - xy$
Factorise.	$2(x^2 - xy) = 1(x^2 - xy)$
Divide both sides by	$(x^2 - xy) \therefore 2 = 1$

[2 marks]

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